

SoundShape — Six Planned Improvements

Emotion-visualizing captions for Deaf and hard-of-hearing viewers · Team Macaroni

1 Haptic Channel — Wristband

Delivers the loudness of the voice as vibration on the wrist.

WHY IT IS NEEDED

Viewers currently have to watch three things at once: the actor's face, the caption text, and the emotion visualization. The visual channel is already saturated. Touch is the only channel that is still completely free — and it requires no eye contact at all.

HOW IT WORKS

- The browser measures loudness from the video audio in real time (Web Audio API)
- Intensity values (0–127) are sent over Bluetooth to the wristband at 25 Hz
- Band: ESP32 + DRV2605L haptic driver + LRA motor (approx. ₩30,000)
- No phone app required — the browser talks to the band directly

Effect. Users feel the rhythm, emphasis, and silences of speech without looking. Freeing the eyes leaves room to actually watch the acting.

2 Low-Confidence Learning Loop

Ask the user only when the AI is uncertain, then learn from those answers.

WHY IT IS NEEDED

The main cause of the 49.6% Korean accuracy is **domain mismatch**. The model learned from AIHub conversational speech, but users watch dramas. Labeled emotion data for Korean drama does not exist anywhere — yet it is exactly what the user is watching right now.

HOW IT WORKS

- When confidence drops below threshold (already shown in gray) → a small prompt appears in the corner, auto-dismissing after 5 seconds
- "How did this scene feel?" with 5–6 options plus "Not sure"
- Rate-limited to once per 2 minutes so viewing is never disrupted
- **Audio is never stored** — only the numeric embedding, which protects privacy
- Once labels accumulate, the SVM is retrained (no GPU needed, takes minutes)

Effect. "How will you improve accuracy?" becomes *"We built a system that improves the more it is used."*

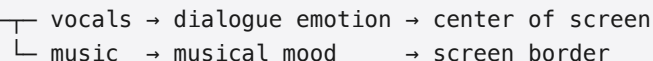
3 Audio Source Separation — One Component, Two Wins

Split voice from background music, then process each independently.

WHY IT IS NEEDED

- Loud background music destabilizes prosody analysis (a limitation we stated ourselves)
- In film and drama, **half the emotion is carried by the score** — and Deaf viewers miss 100% of it

HOW IT WORKS

audio — separate — 

- **The two tracks are never merged** — they stay independent
- Music is processed in coarse 5–10 second windows (music changes slowly)
- Videos without music skip separation entirely, preserving speed

Effect. Accuracy holds up in noisy scenes, and the emotional information in the score finally reaches the viewer. When dialogue and music diverge — cheerful words over an ominous cue — that contrast is itself the director's intent, so it must be shown separately rather than averaged away.

4 Speaker Separation — Whose Emotion Is It?

When several people speak, show each person's emotion separately.

WHY IT IS NEEDED

Drama is mostly conversation between two or more people. Emotion is relational — "**A is angry at B**" is the substance of a scene, but right now only "someone is angry" comes through.

HOW IT WORKS

- Extract a speaker signature per utterance, then cluster similar ones (works in real time)
- The number of speakers does not need to be known in advance
- **Distinguish by position, not color** — A on the left, B on the right

Why not color? Color already encodes emotion (red = anger). Using it for speaker identity too would make the two signals collide and neither would be readable.

Effect. Multi-person scenes become legible, and relational emotion comes through. This mirrors *role shift* in sign language interpreting, where interpreters shift body position to indicate different characters — so the design has precedent.

5 Hybrid Rendering — Measure What Can Be Measured

Only color comes from the AI's emotion judgment; motion, thickness, brightness, and speed come directly from the actual voice.

WHY IT IS NEEDED

All of the current error originates in the classification step. But **the prosody measurements themselves are never wrong** — an F0 of 187 Hz is a fact. Conversely, rendering from raw prosody alone would lose the emotion category entirely: excited joy and anger look identical in motion. So the boundary is drawn where it belongs — **measurement stays measurement, and only inference goes to the AI.**

HOW IT WORKS

Visual channel	Source	Error
Color (hue)	AI emotion classification	Present
Motion, thickness, brightness, speed, tremor	Measured prosody	None

This split is supported by our own data: **arousal is measured reliably from acoustics ($r = 0.73$), while valence is weak ($r = 0.42$).** The motion-family channels are arousal-driven, so measurement is accurate there; color is valence- and category-driven, so it genuinely requires inference.

FOLLOW-UP EXPERIMENT

Compare **full-AI** against **hybrid** rendering on identical clips. Hybrid may well win — a trembling voice would tremble on screen by exactly that amount, rather than being flattened into a category-based motion preset.

Effect. "But half of it is wrong" becomes *"Only one channel — color — can be wrong. The other four are a direct transcription of the actual voice, so they carry no error."*

6 Personal Calibration — Available On Demand

Users who want to can run a short test in settings, and the channels that work best for them are strengthened.

WHY IT IS NEEDED

Deaf and hard-of-hearing people are not a uniform group. Residual hearing, age of onset, and sign-language use all differ, and **some people read color well while others read motion well.** There is also a question we currently cannot answer: "Have you verified that the mapping is intuitive?"

HOW IT WORKS

- **Not forced at first launch** — the research-based defaults work immediately
- Settings → "Customize visual expression" → a 2-minute test, repeatable at any time

- The test shows a visualization and asks "Which emotion is this?" 12–15 times, **measuring each channel separately**
- Manual sliders are available for users who prefer to adjust without the test

Effect. An unverified weakness turns into a feature: "*We measure how intuitive the mapping is for each individual user and adapt to them.*" The calibration data from those users then becomes research evidence on mapping intuitiveness in its own right.

Overview

#	Improvement	Problem it solves	Difficulty
1	Haptic channel	Visual channel saturation	Medium (hardware)
2	Low-confidence learning loop	49.6% accuracy / domain mismatch	Low
3	Audio source separation	Noise sensitivity + missing score	Medium–High
4	Speaker separation	Multi-person scenes collapse	Medium
5	Hybrid rendering	Classification error	Medium
6	Personal calibration	Unverified mapping intuitiveness	Low

How They Reinforce Each Other

- **3 → 1** — With separation, the haptics follow *voice* loudness only, not the background music
- **3 → 5** — Cleaner vocals mean more accurate prosody, which directly sharpens hybrid rendering
- **4 → 1** — With two motors, each speaker can vibrate at a different location (left wrist = A, right = B)
- **2 ↔ 6** — Both ask the user about emotion, so they *share the same UI component*
- **5 ↔ 6** — Calibration can also decide whether a given user is better served by more AI color or more raw prosody

Suggested Order of Work

When	Task
Immediately	#2 survey UI · order hardware parts (data takes time to accumulate; shipping takes 1–2 weeks)
Weeks 1–2	#5 hybrid rendering · #4 speaker separation
Weeks 2–3	#1 haptic assembly · #6 calibration (shares UI with #2)
Weeks 3–4	#3 source separation (heaviest, most performance risk)
Final	#2 retraining · #5 comparison experiment